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RESEARCH ARTICLE



Effectiveness of video-based modelling to facilitate conversational turn taking of adolescents with autism spectrum disorder who use AAC

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ABSTRACT

This study evaluated and compared the effectiveness of packaged video modelling (VM) and video self-modelling (VSM) interventions to develop conversational behaviors with four adolescents with autism spectrum disorder (ASD) who used augmentative and alternative communication (AAC). The study was conducted using an alternating treatments design nested within a multiple baseline design. The intervention effect was measured using Robust-Improvement Rate Difference (R-IRD). The results demonstrated that, overall, video-based modelling used in conjunction with a system of least prompts was effective in promoting conversation skills in adolescents with ASD who used AAC. Without the systematic instruction, R-IRD indicated that these techniques yielded only small to moderate intervention effects. The findings demonstrated the necessity of systematic instruction for this group of participants. This investigation provides preliminary evidence to support the use of packaged video-based modelling interventions to develop conversation skills in adolescents with ASD who use AAC systems.

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Impairments in social communication skills are universal across individuals with autism spectrum disorder (ASD) (American Psychiatric Association, 2013), regardless of their age or cognitive ability (e.g., Howlin, 2005). Individuals with neuro-typical development acquire critical social communication skills naturally through everyday interaction with the people around them; however, individuals with ASD are less likely to develop these skills without explicit instruction (Bellini, 2007). Therefore, social communication skills are expected targets in interventions for these individuals (Bellini & Akullian, 2007).

Video modelling (VM) is a type of video-based intervention (VBI) that has emerged as an evidence-based practice for individuals with ASD (Wong et al., 2015), and is founded on the Social Learning Theory (Bandura, 1971). There are many types of VBI, including VM (e.g., Charlop & Milstein, 1989), video self-modelling (VSM) (e.g., Sherer et al., 2001), both of which are presented from the vantage point of the third person; and point-of-view modelling (PoVM; Tetreault & Lerman, 2010), which is presented from the vantage point of the learner. In video-based interventions, the learner watches target behaviors demonstrated by a video clip, with the expectation that these behaviors will be learned and later reproduced. The model demonstrating the target behavior in the video clips could be an adult (e.g., Charlop & Milstein, 1989), a peer (e.g., Ogilvie, 2008), an animation character (Ogletree, Fischer, & Sprous, 1995), or the learner self, as in VSM (Dowrick, 1999).

The use of VBI in facilitating social communication skills has been researched extensively, and the general consensus

is that it is effective in promoting a range of skills in individuals with ASD (e.g., Bellini & Akullian, 2007; Mason et al., 2013; Shukla-Mehta, Miller, & Callahan, 2010). The majority of these studies have focused on discrete communication components such as responding (e.g., Ogletree et al., 1995), requesting (e.g., Banda, Copple, Koul, Sancibrian, & Bogenschutz, 2010; Copple, Koul, Banda, & Frye, 2015), and greeting (Simpson, Langone, & Ayres, 2004). Rogers (2000) conducted a review of interventions that facilitate social skills in children with ASD and found that interventions targeting social interactive skills indirectly facilitated the development of other discrete communication and behavioral skills. Conversation is the key component of social interaction (Goodwin & Heritage, 1990). To date, two reviews have specifically examined the use of VBI to develop conversation skills in individuals with ASD (Sng, Carter, & Stephenson, 2014; Thirumanickam, 2017).

The review by Sng et al. (2014) included 13 studies and 29 participants aged from 4- to 11-years-old. The aim of this review was to determine (a) the overall effectiveness of VBI in facilitating conversation skills, in particular the “verbal exchange of social information” (p. 111); and (b) differential intervention effects between the different types of video models. The authors included studies that targeted social initiations such as greetings and excluded studies that targeted communicative initiations that had an instrumental or behavior-regulating purpose (e.g., requesting). The findings showed that VBI was somewhat effective in developing conversation behaviors for individuals with ASD and that there were only

small differences between the types of models employed; however PoVM, used in one study (Tetreault & Lerman, 2010), was ineffective.

The review by Thirumanickam (2017) looked at 24 participants between 6 and 26-years-old from 11 studies that targeted conversational turn taking. Studies in which discrete components of social communication were targeted, such as initiations (interactional or instrumental), responding, and complementing, were excluded from the review. The findings showed that VM was moderately effective, however, it seemed premature to draw any conclusion concerning the evidence-base for VSM and PoVM because these methods were only used in one study respectively. In the review, Thirumanickam highlighted four issues. First, the study by Sherer et al. (2001) was the only one that compared the effectiveness of peer and self video-models to facilitate the conversation behavior of individuals with ASD. They found both VM and VSM to be equally effective in facilitating the acquisition of the target skill. Second, adolescent and adult participants with ASD were few in number, a limitation also pointed out by Sng et al. (2014). Third, the majority of VBI involving adolescent participants addressed activities of daily living as goals such as meal preparation, purchasing items, using the bank machine, and working the microwave (e.g., Smith et al., 2016). The few studies that focused on social skills were aimed at discrete components of social communication such as social initiations (Nikopoulos & Keenan, 2003) and spontaneous requesting (Banda et al., 2010). Fourth, none of the reviewed studies included individuals with ASD and complex communication needs, although it has been estimated that about 30% of these individuals require the use of AAC (Ganz, 2015; Mirenda & Iacono, 2009).

The few investigations into the use of VBI to facilitate communication skill development in individuals with ASD and complex communication needs targeted requesting skills. In these studies, VM was implemented either exclusively or in conjunction with additional instruction (Banda et al., 2010; Cihak, Smith, Cornett, & Coleman, 2012; Copple et al., 2015). Prompting and prompt-fading techniques have been common additions to VBI because they have been found to be effective in facilitating communication skill development (Goldstein, 2002; MacDuff, Krantz, & McClannahan, 2001). Prompting has also been listed as an effective instructional strategy for individuals with ASD (Wong et al., 2015). Nonetheless, research that combined prompting with VBI rarely used prompting systems as the primary intervention method.

Banda et al. (2010) and Copple et al. (2015) targeted spontaneous requesting skills using VM in children with ASD who used speech-generating devices. Two participants, aged 17 and 21, engaged in the study by Banda et al. and three pre-school children were included in the study by Copple et al. Banda et al. presented the VM intervention without additional prompts, while Copple and peers supplemented the video modelling with least-to-most prompting in a replication study. None of the participants in these studies demonstrated generalization of the learned task across stimuli. The authors of both studies posited that including

prompting in the intervention package might have enhanced the acquisition of the target skills. Cihak et al. (2012) evaluated the effectiveness of VM in combination with Picture Exchange Communication System (PECS) (Bondy & Frost, 1994) instruction to teach independent requesting to four 3-year-olds with ASD. The results showed that the packaged intervention (VM plus PECS) was more effective than when PECS was implemented alone. Consistent with the findings of recent reviews of AAC research (Ganz 2015; Holyfield, Drager, Kremkow, & Light, 2017), requesting was the main communication behavior targeted in VBI studies for participants who use AAC (Banda et al., 2010; Cihak et al., 2012; Copple et al., 2015). While requesting is an important early communication skill, the communication goals for adolescents and adults with ASD who use AAC should move beyond the expression of needs and wants, to increasing social closeness and improving information transfer (Light, 1988) in order to meet changing needs and experiences (Holyfield et al., 2017).

The purpose of the current study was to evaluate and compare the effectiveness of packaged video modelling and video self-modelling interventions to develop conversational behaviors with adolescents on the autism spectrum, who used augmentative and alternative communication. The research questions were: (a) what was the effectiveness of VM and VSM, when implemented with the system of least prompts in developing conversational behaviors in adolescents with ASD, who use AAC? and (b) were there any differential effects on the target behaviors based on the type of video model?

For the purpose of this investigation, "conversation interaction" was defined as an act of responding to the preceding question and asking a follow-up question to maintain the interaction. A packaged intervention was implemented because previous research had demonstrated that participants with intellectual disabilities require additional error-correction procedures to improve their overall learning via video modelling (Banda et al., 2010; Copple et al., 2015). All participants in the current study had intellectual disabilities. First-person perspective (PoVM) was not included because this technique might not be consistent with the cultural conventions of video viewing (Sng et al., 2014).

Method

Participants

Participants were recruited from a range of special schools in South Australia. Ethical approval for the study was obtained from Flinders University, the education departments of public and catholic schools, as well as the professional autism state organization. Four participants (two across each of two strands of the study) were recruited. All met the following inclusion criteria: (a) were aged between 11;0 and 18;0 (years; months); (b) had a diagnosis of ASD, in accordance with DSM-IV (DSM-V was released after the call for participants was advertised); (c) were deemed by their classroom teachers to be context-dependent communicators, that is, able to use symbolic or non-symbolic modes to express a limited range of

Table 1. Characteristics of participants.

Age/Gender	Ethnicity	Language age ^a	Communication characteristics	Sensory/behavior characteristics	Interests
Sam 11;2 (M)	Caucasian	6;3	No speech; uses gestures and finger-traces letters of the alphabet; able to construct at least three-word sentences using his device to express request and comment; uses Proloquo2Go with intermediate core vocabulary ^c	None observed or reported	Interacting with others, especially around preferred topics like Teletubbies ^e
Lily 11;4 (F)	Caucasian	<2;0 ^b	Can produce one-word utterance using speech; unintelligible speech; able to construct three-word sentence to request desired item using her AAC; uses Proloquo2Go with customized vocabulary ^d	Reported sleep difficulty (often appears tired); hand-flapping and rocking behaviors, chews on chewy tubes or clothing	Enjoys watching cartoons, especially Peppa Pig, ^f and playing by herself
Dan 18;0 (M)	Vietnamese	2;3	Echolalic speech; speech always starts with “I want” and ends with “please”; able to construct three-word sentence to request desired item using his AAC; uses Proloquo2Go with customized vocabulary	Self-harming behaviors (picks on his skin and bruises); humming and rocking behaviors	Enjoys colours and iPad games
Pete 11;0 (M)	Vietnamese	2;5	Echolalic speech; likes typing on the keyboard function on the AAC system; uses Proloquo2Go with intermediate core vocabulary	Screaming and hitting behaviors	Enjoys iPad games, The Wiggles ^g

Notes. Language age was derived using the standard scores in the Peabody Picture Vocabulary Test (4th ed.) (PPVT-4B), Dunn and Dunn (2007); self-recognition score refers to the participant's ability to discriminate self from others in photographs.

^aAge (year; months).

^bLily's standard score did not yield any age equivalent score.

^cIntermediate core vocabulary is an in-built vocabulary system within the Proloquo2Go, which supports full language development.

^dCustomized vocabulary was developed on a blank template and was limited in its communicative functions.

^eTeletubbies is a British preschool children's television series created by Ragdoll Productions' Anne Wood and Andrew Davenport.

^fPeppa Pig is a British pre-school animated television series.

^gThe Wiggles are an Australian children's music group.

communication functions in limited contexts (Blackstone, 1999); (d) showed an interest in watching videos and had no objections to being filmed; and (e) attended special schools, for which the eligibility for enrolment in South Australia is based on the presence of intellectual disabilities. All of the participants in this investigation were ambulant and able to communicate simple requests (e.g., I want ...) using their speech-generating device (SGD). Participant characteristics are presented in Table 1. Pseudonyms are used throughout to protect confidentiality. All participants used Proloquo2Go.¹ The details of the vocabulary template used are provided in Table 1.

Design

An alternating treatments design (Barlow & Hayes, 1979) embedded within a multiple baseline design (Kazdin &

Kopel, 1975) was used to address the research questions. This design was used to compare the effectiveness of VM and VSM in facilitating the development of the conversation behaviors. The order in which each intervention was applied was alternated and counter-balanced across the conditions to control for order effects (Kratichwill & Levin, 2014). Alternating and counter-balancing the conditions are critical elements of this design (Kazdin, 2011). A known limitation of the design is the likelihood of a carryover effect from one intervention to the next (Kazdin, 2011). Nonetheless, the aim of this investigation was first to determine whether VM and VSM were effective in facilitating the development of the target behavior, before comparing the differential effects between the two interventions.

The essential feature of the multiple baseline design is the successive introduction of the intervention across the data series (participants, settings, behavior). This means that intervention is not introduced to all series simultaneously; instead intervention for subsequent data series is introduced following observable changes in the preceding data series.

¹Proloquo2Go is a trademark of AssistiveWare B.V., Amsterdam, The Netherlands, <http://www.assistiveware.com/product/proloquo2go>

This is to control for the possibility that any changes in the dependent variable measures are linked to the independent variable and not potentially due to other uncontrolled events (e.g., age, practice, maturity) (Kazdin, 2011). In some cases, a longer lag between the introduction of the intervention and the observable change can lead to prolonged baselines for the subsequent series. Prolonged baselines have been found to reduce methodological rigour (Kazdin, 2011; Kratochwill & Levin, 2014) and might increase irrelevant and contesting behaviors in participants (Panyan, Boozer, & Morris, 1970). To counter-act the potential negative effects of prolonged baselines, a pre-determined intervention start point system was used to ensure the interventions were introduced in a successive pattern, in line with the requirements of the multiple baseline design. Due to the small number of participants (two) initially recruited, a second strand of the study was conducted to ensure adequate replication of effect. This second strand also involved two participants.

Measures

Independent variables. The independent variables were (a) VSM with a system of least prompts, and (b) VM with a system of least prompts. Prompts were presented in the least-to-most hierarchy (i.e., verbal, navigational, gestural, modelling, and physical).

System of least prompts. A six-tier prompting system was used in a least-to-most hierarchy to elicit target behavior during the intervention sessions. This method of prompting offers a systematic way to fade prompts and promote independent performance for many learners with ASD (Libby, Weiss, Bancroft, & Ahearn, 2008; MacDuff et al., 2001), and allows for more independent responses before offering more intrusive prompts (Collins, 2012). Moreover, the system of least prompts produces more rapid skill acquisition than a most-to-least prompting system (Libby et al., 2008; MacDuff et al., 2001). The hierarchy of prompts in this study included repetition of a preceding question (R), followed by verbal (V), navigational (N), gestural (G), modelling (M), and physical (P) prompts. See Appendix A (Supplemental materials) for operational definitions for each level and their associated scores. The first author waited 5–10 s for the participant to demonstrate the target behavior independently. If an independent response was not observed within 10 s, then the researcher provided the first level of prompt and waited 5–10 s. A 5–10 s latency was offered between each level of prompt.

Dependent variables. Intervention effectiveness was measured based on (a) percentage of target conversation behaviors (responding to preceding question and asking a follow-up question), and (b) type and number of prompts required. Conversational interactions followed the pattern in the conversation scripts as shown in Appendix B (Supplemental materials). The maximum score for each interaction was 12 points: six for independently answering the preceding question and six for independently asking a follow-up question.

Materials

Conversation scripts. Given that previous research showed that VM did not yield any differential effects for conversational initiations in preferred and non-preferred topics (Haymes, 1995), this study used generic topics to engage school-aged individuals between the ages of 10 and 18 years: Food, Holiday, Movie, Colours, Recess, and Weekend. Open questions such as “What about you?” and “What’s yours?” were used instead of script-specific questions such as “What did you eat for breakfast?”. The use of generic questions was also deemed to improve the generalizability of the targeted skills. Four scripts were used to teach turn-taking skills – two each for VM and VSM – and four additional scripts were used to measure generalization. Each script had one of two turns: “What’s yours?” or “What about you?”. The same scripts were used for all participants, and only the answer to the initial question in each script was unique to each participant. These unique answers had been elicited from the participants or their parents and teachers in advance of video production. For an example of a conversational interaction, see Appendix B (Supplemental materials).

Video vignette of exemplary interaction. Four video clips, two each for VSM and VM, were developed. The VSM clips involved the participant, while the VM clips involved a gender- and age-matched peer as the primary model. The male peer model was 11-years-old and the female peer model was 13-years-old; both were Caucasian. The first author acted as the communication partner in all video clips, each of which was approximately 6-s long. All video recordings were produced using the Sony HDR-C240 video camera and were edited using Sony Play Memories.² Examples of conversational interactions are provided in Appendix B (Supplemental materials).

The video clips for VM and VSM were recorded and developed prior to the baseline phase. The VM clips were recorded in a meeting room at the authors’ university. The peer models’ answers and follow-up questions were pre-stored in the communication application and rehearsed with the researcher prior to recording. The overall duration for producing the VM clips, including recording and editing, was approximately 30 min per script, per peer.

The VSM clips were recorded in school-appointed areas. The feed-forward method for VSM was used because the intervention involved teaching new skills that were not in the participants’ repertoire. The researcher prompted the participant to navigate the SGD and locate the answer to a question as well as the follow-up question required for the interaction. The instructions used during the video recording were navigational and gestural prompts. For navigational prompts, the participant was verbally directed to the appropriate folders or categories (e.g., *Go to the home page*). For gestural prompts, the researcher pointed to the preferred folder/category or symbol while saying, for example, *Press this*. The prompts were removed during final editing,

²Sony HDR-C240 is a handycam, camcorder and Sony Play Memories is a media storage and editing software developed by Sony, Tokyo, Japan, <https://www.sony.com.au/>

resulting in a scene of the researcher and the participant engaging in a continuous conversational interaction. The duration for producing the VSM clips was approximately 90 min per script.

Procedure

Setting. All assessment and intervention sessions were conducted by the first author in a separate room at the special school attended by the participants, away from the classroom.

Assessments. The Peabody Picture Vocabulary Test-4 (PPVT-4) (Dunn & Dunn, 2007) was administered to assess the level of receptive vocabulary. A self-recognition assessment was used on two separate occasions to ensure that each participant was able to distinguish self from others. This was particularly important when comparing the differential effects between peer- versus self-models in the video clips. During the self-recognition assessment, the participants were required to identify themselves and their classmates in different sets of photographs offered by their classroom teacher. Scoring was based on whether the participant identified the right or wrong photographs. All participants achieved 100% success, without any errors. All the assessments were conducted by the first author.

Baseline. The two participants in each strand of the study commenced baseline simultaneously. Following five baseline sessions the first participant commenced intervention, while the second participant remained in baseline for up to four baseline probes, during which the participants were assessed on their interaction during the eight scripted conversations (four each for intervention and generalization). The researcher initiated the conversation with a question and waited for the participant to respond. If the participant did not respond, the same question was repeated once and if there was still no response, the researcher provided her response and terminated the interaction. The participants did not receive any intervention in baseline. The performance was scored based on the percentage of conversational behavior (six points for an independent response or 0 for no response) recorded.

Intervention. Intervention was conducted between three and four times per week, for up to nine sessions. During each session, participants were shown the video clips, one at a time. After each viewing, the participant and the researcher engaged in the same scripted conversation interaction presented in the video clip. The participants were allowed a 5–10-s latency interval to demonstrate the target behavior independently and if they failed to do so, prompts were provided based on the system of least prompts. The score for conversational behavior in the intervention phase was based on the prompt that elicited the target response (Appendix A). For example, if participants answered a preceding question independently but required a navigational prompt to ask the follow-up question, they received six points for independently answering the question and three points for the navigational prompt that elicited the target follow-up question. The maximum score for each interaction was 12 points

(six points for independently answering the preceding question and six points for independently asking a follow-up question). Raw scores were converted to percentages of scripted conversational behavior (e.g., $9/12 = 75\%$). Each training trial was followed by a short break before presenting the next video clip. Participants watched four videos in each session (two different scripts each for VM and VSM).

Generalization and maintenance. Four separate untrained conversation scripts were used to measure generalization during baseline, intervention, and maintenance phases. Maintenance of skills was assessed at least 2 weeks post intervention, using all scripts involved in intervention and generalization. The participants did not receive any intervention in the generalization or maintenance phases. The performance was scored based on the percentage of conversational behavior demonstrated (six points for an independent response or 0 for no response).

Data analysis

Effect size. The Robust-Improvement Rate Difference (R-IRD) was used to calculate the effect size (Parker, Vannest, & Brown, 2009). R-IRD scores range from -1.0 to 1.0 . An IRD score of 1.0 means maximum improvement from baseline to intervention, whereby no points in the baseline and intervention phases overlapped. There is no established standard for IRD/R-IRD; however, a tentative benchmark is set as follows: (a) <0.5 = ineffective/questionable, (b) between 0.51 and 0.70 = a moderate effect, and (c) >0.71 = a large effect (Parker et al., 2009).

Inter-observer reliability (IOR). The first and fourth author independently observed and recorded the performance for at least 20% of sessions for all participants and phases (baseline, intervention, generalization, and maintenance), based on a pre-determined scoring scheme. Any disagreement regarding the performance score was discussed and resolved. The average IOR for prompted conversation behavior was 84% (range: 50–100%), which is regarded as an acceptable agreement rating (Kazdin, 2011).

Treatment fidelity. Using an eight-step checklist, an independent observer evaluated the procedures undertaken during the baseline and intervention phases for 20% of sessions for all participants. The checklist included adherence to the prompting protocol (level of prompts, latency between each prompt) and viewing of video clips. The average treatment fidelity was 93.8% (range: 63–100%).

Social validity

Using the social evaluation method (Kazdin, 2011), pre- and post-intervention video clips, including performance during maintenance, were shown to participants' teachers to verify identifiable changes in participant conversational behaviors. The teachers viewed one baseline, two intervention, and one maintenance clip for each participant. Lily's teacher watched one baseline and two intervention clips, as maintenance was not conducted for Lily. The teachers were not informed of the chronological order of each video clip (i.e., whether the

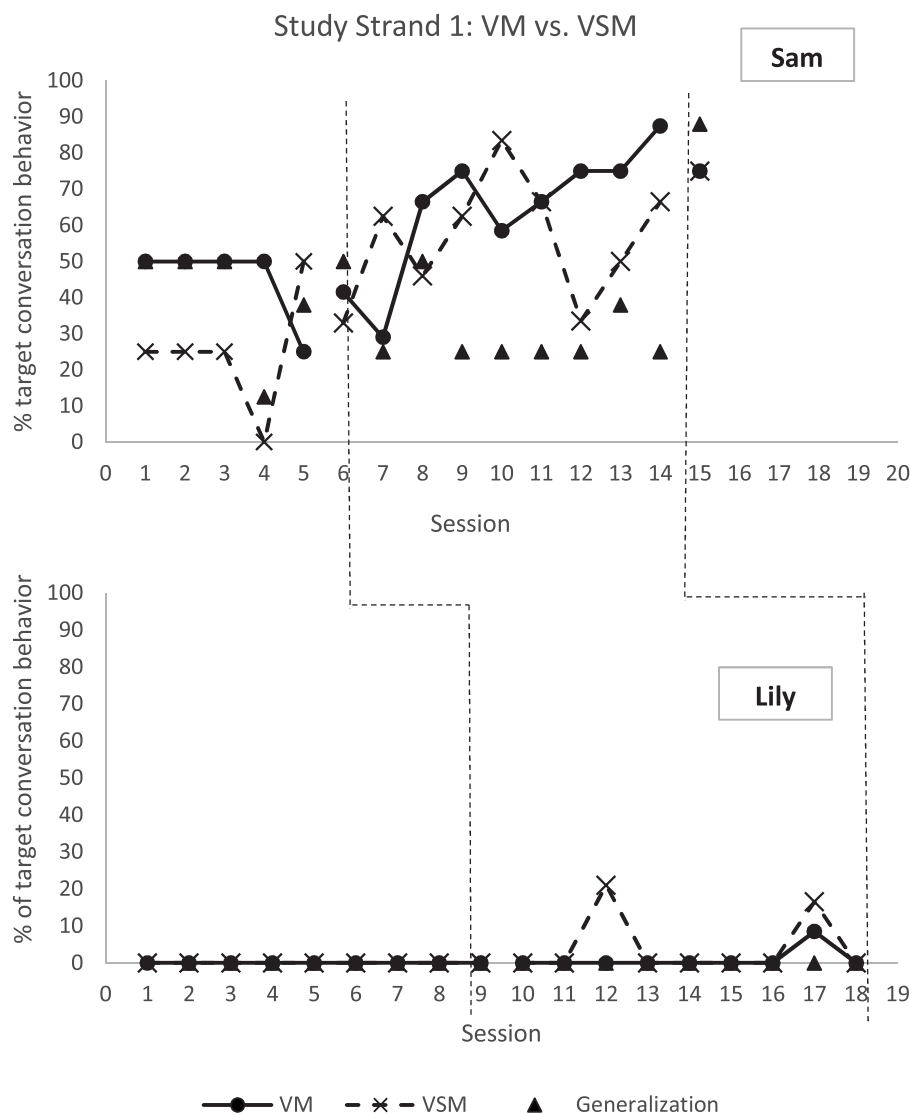


Figure 1. Average percentage of target conversation behavior for Sam and Lily, for baseline, intervention, maintenance, and generalization. Interaction during baseline and maintenance did not include any instruction (VM or VSM or system of least prompts).

clips were recorded during baseline, intervention, or maintenance). They completed an eight-question survey to rate the frequency and quality of conversational interactions on a five-point scale, ranging from 1 (*Never*) to 5 (*Always/Fluently*), after watching the 10–15-min video clips. They also provided additional comments regarding the benefits they perceived following the intervention.

Results

Overall, three out of the four participants demonstrated improved performance following the implementation of the packaged intervention. Figures 1 and 2 display the percentage of prompted conversational behavior for the first and second strands of the study, respectively. Table 2 displays the intervention effect size for the target behaviors, including the upper and lower parameters for the adjusted confidence intervals at 87.89% (Payton, Miller, & Raun, 2000). The p -values were obtained from the Fisher Exact Test.

Video modelling (VM). For three out of the four participants, VM had moderate to high intervention effects on the

target behavior. The effect size for both independent and prompted conversation behaviors during the VM intervention for Scripts 1 and 2 is shown in Table 3. Sam's performance following the packaged VM intervention showed a moderate effect ($R-IRD = 0.69$), and the results were statistically significant ($p = 0.02$). Lily's conversational behaviors did not change. Although Dan's and Pete's overall performances in the VM scripts were variable, there was an overall increasing trend in intervention for both. It is important to note that neither participant demonstrated independent turn taking during baseline.

Video self-modelling (VSM). VSM had variable effects for the four participants. The packaged VSM intervention showed a large effect ($IRD = 0.84$) and the results were statistically significant ($p < 0.001$) for Sam. Lily's conversational behaviors did not change. There was a fluctuating trend in Dan's performance following the introduction of the packaged VSM intervention, which demonstrated very large effects on Pete's performance – $IRD = 1.00$ [0.55, 1.00] – because there was no overlap between his performance in conversation behavior during baseline versus intervention

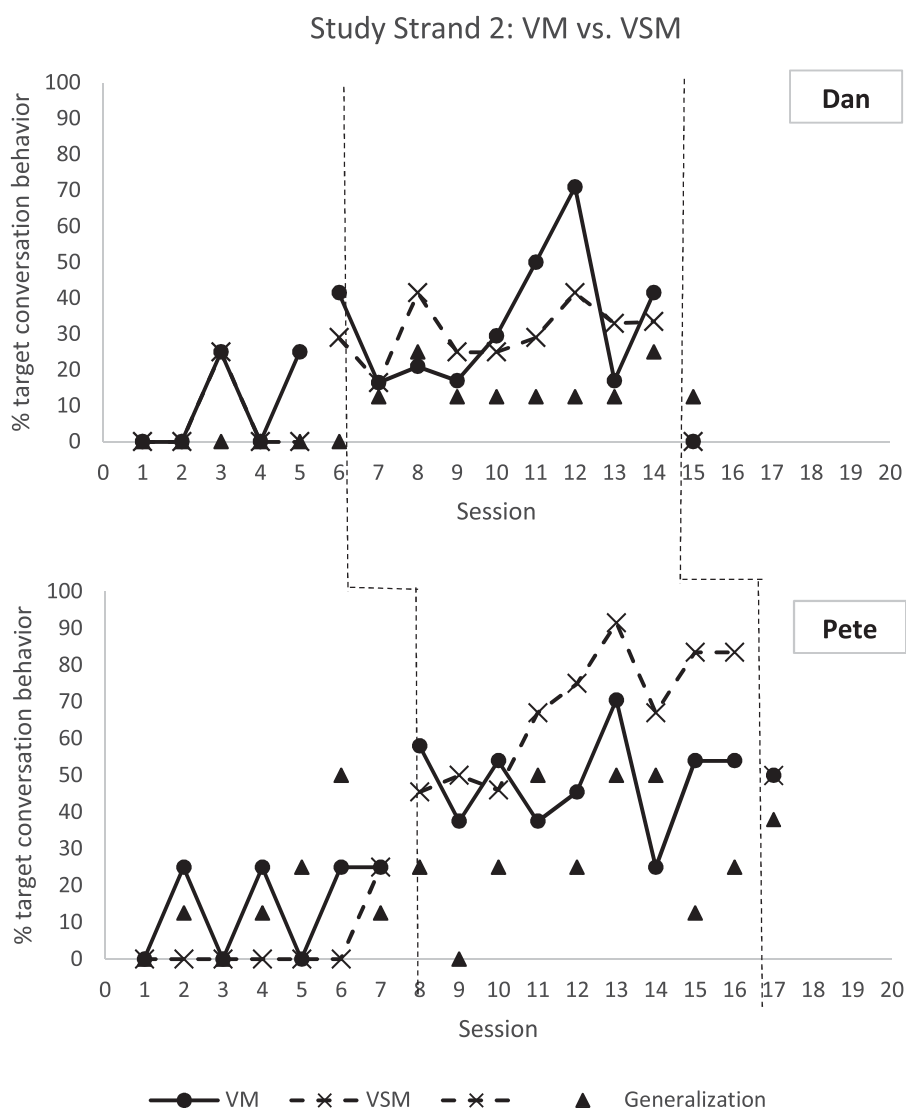


Figure 2. Average percentage of target conversation behavior for Pete and Dan, for baseline, intervention, maintenance, and generalization. Interaction during baseline and maintenance did not include any instruction (VM or VSM or system of least prompts).

Table 2. Robust-Improvement Rate Difference (R-IRD) for the packaged video modelling and video self-modelling interventions.

Participant	Packaged video modelling				Packaged video self-modelling			
	<i>p</i>	R-IRD	87.89% CI		<i>p</i>	R-IRD	87.89% CI	
Sam	0.02	0.69	[0.12, 0.96]		>0.001	0.84	[0.28, 1.00]	
Lily	1.00	0.11	[-0.34, 0.53]		0.22	0.35	[-0.24, 0.62]	
Dan	0.02	0.69	[0.12, 0.96]		0.00	0.84	[0.28, 1.00]	
Pete	>0.001	0.87	[0.39, 1.00]		>0.001	1.00	[0.55, 1.00]	

Note. The Robust-Improvement Rate Difference (R-IRD) is from Parker et al. (2009); CI = confidence interval.

(see Figure 2). The results were statistically significant ($p < 0.001$) (see Table 4). Table 4 displays the effect size for both independent and prompted conversation behaviors during the VSM intervention for Scripts 1 and 2.

Level and number of prompts. There was a decline in the level of prompts from intervention sessions one to nine, for Sam, Dan, and Pete. There was variation in the number and type of prompts required by these participants throughout intervention. Sam and Pete demonstrated more independent responses as the intervention progressed, while Dan continued to require modelling prompts throughout the

intervention, as seen in Figure 3. Lily continued to require full-physical prompts throughout the intervention.

Generalization. One out of the four participants (Pete) demonstrated conversational behaviors in the generalization scripts. Sam initiated new topics rather than taking turns in the same topic to maintain the interaction, while Dan responded only to the researcher's questions, but did not ask the follow-up questions. Lily did not respond or ask follow-up questions during the generalization probes.

Maintenance. A maintenance effect was observed for Sam and Pete, although there was a decline in performance following intervention. As previously noted, maintenance could not be established for Lily, and Dan did not maintain the intervention effects during the maintenance phase.

Social validity. Overall, the teachers reported that the frequency and quality of interaction increased after the intervention for Sam, Dan, and Pete. Lily's classroom teacher reported that, as the sessions progressed, Lily's communication device use for initiation increased for requesting objects of interest but not for participating in the scripted interaction. All teachers commented that the participants'

Table 3. Robust-Improvement Rate Difference (R-IRD) for the independent and prompted interaction for video modelling (VM).

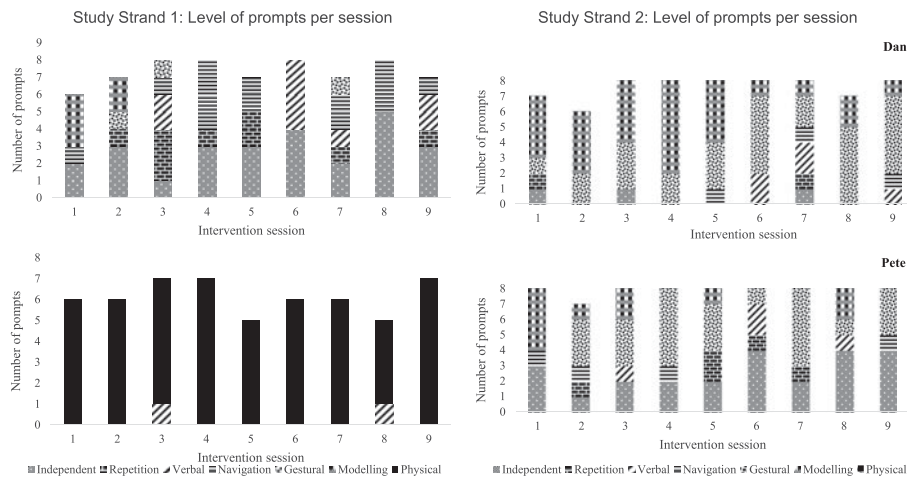
Participant	Script 1						Script 2					
	Independent			Prompted			Independent			Prompted		
	<i>p</i>	R-IRD	87.89% CI	<i>p</i>	R-IRD	87.89% CI	<i>p</i>	R-IRD	87.89% CI	<i>p</i>	R-IRD	87.89% CI
Sam	0.09	−0.40	[−0.56, 0.17]	0.09	0.53	[−0.03, 0.89]	1.00	0.07	[−0.39, 0.58]	0.09	0.53	[−0.03, 0.89]
Dan	0.58	−0.24	[−0.53, 0.37]	>0.0001	1.00	[0.46, 1.00]	>0.001	−0.71	[−0.56, 0.16]	0.02	0.69	[0.12, 0.96]
Pete	1.00	−0.02	[−0.47, 0.46]	>0.0001	0.87	[0.39, 1.00]	1.00	−0.02	[−0.47, 0.46]	0.01	0.62	[0.11, 0.91]

Note. The Robust-Improvement Rate Difference (R-IRD) is from Parker et al. (2009); CI = confidence interval.

Table 4. Robust-Improvement Rate Difference (R-IRD) for the independent and prompted interaction for video self-modelling (VSM).

Participant	Script 1						Script 2					
	Independent			Prompted			Independent			Prompted		
	<i>p</i>	R-IRD	87.89% CI	<i>p</i>	R-IRD	87.89% CI	<i>p</i>	R-IRD	87.89% CI	<i>p</i>	R-IRD	87.89% CI
Sam	0.27	0.38	[−0.17, 0.80]	0.02	0.69	[0.12, 0.96]	0.58	−0.24	[−0.53, 0.32]	0.02	0.69	[0.12, 0.96]
Dan	0.09	−0.40	[−0.56, 0.17]	>0.001	0.84	[0.28, 1.00]	1.00	>0.001	[0.00, 0.00]	>0.001	1.00	[0.46, 1.00]
Pete	0.09	0.62	[0.11, 0.91]	>0.001	1.00	[0.55, 1.00]	>0.001	0.87	[0.39, 1.00]	>0.001	1.00	[0.55, 1.00]

Note. The Robust-Improvement Rate Difference (R-IRD) is from Parker et al. (2009); CI = confidence interval.

**Figure 3.** Type and number of prompts per intervention session for all participants.

keenness to explore and use the devices for communicative purposes increased following participation in the intervention. Pete's teacher reported that Pete's word approximations had improved since participating in the intervention.

Discussion

The overall findings from this study showed positive treatment effects for developing conversational behavior in three out of four participants, following an intervention structured around video modelling plus prompting. The three participants who demonstrated positive treatment effects showed varying levels of generalization across scripts. Maintenance effects were noted for Sam and Pete, who demonstrated conversational behaviors above their baseline level. The decline in performance from intervention to maintenance might be attributed to the lack of meaningful opportunities to rehearse learned skills outside of the intervention setting as the long-term retention of behaviors is aided by rehearsal opportunities (Bandura, 1971). The lack of maintenance for

two participants (Lily and Dan) was not surprising, given that additional prompts and reinforcement were not part of the maintenance phase, and they demonstrated limited acquisition of the target skills before the maintenance phase. Lily only engaged to request desired items, and Dan was dependent on cues for interaction. For this group of participants, a longer intervention phase might have been required to achieve adequate levels of skill acquisition and fluency.

Between- and within-participant variability

The findings of the study indicate that the success of learning via video medium might not have been influenced by the learners' language ability, consistent with the postulations of Sherer et al. (2001). Sam and Dan demonstrated the same levels of success, despite Sam's more advanced language age (see Table 1). Meanwhile, Pete showed higher success in both interventions, although his language abilities were equivalent to Dan's and lower than Sam's (see Table 1).

Performances across scripts were variable across three of the four participants. One explanation for this variability might be a personal preference for topics (e.g., Dan performed better in food-related topics). Haymes (1995), for example, found no differential effects for obsessive versus non-obsessive conversation topics in promoting verbal initiations in children with ASD. Conversely, Copple et al. (2015), found that the use of preferred items augmented treatment outcomes for spontaneous requesting. Topic preference was not part of the current investigation, thus, this issue remains unresolved.

Differential effects between peer and self as video models

Consistent with the findings of previous studies (e.g., Sherer et al., 2001), both peer and self-video models were effective in promoting the target skill acquisition for one participant (Pete), but the self-model demonstrated slightly higher effects than the peer model for two participants (Sam and Dan). Mason et al. (2013) found that while adult models demonstrated highest treatment effect ($IRD = 0.87$), self-models ($IRD = 0.81$) demonstrated slighter greater treatment effect than peer models ($IRD = 0.70$). The lack of treatment effects for Lily, despite being engaged in the video clips (and especially the VSM clips as observed during intervention sessions), might have been influenced by a special interest in videos and in her own image. Shukla-Mehta et al. (2010) suggested that VSM might be counterproductive for individuals who are more fascinated by their image than the intended targeted behavior.

The importance of systematic instruction (SI)

Findings from this investigation show that added instruction was imperative in enhancing treatment effects, consistent with the findings from previous studies that implemented VM to individuals with ASD who use AAC (Banda et al., 2010; Copple et al., 2015). Although the number and intensity of prompts in the current study was reduced as the intervention progressed, all participants required a prompting system to engage throughout the study. Previous reviews suggested that additional instruction could increase cognitive burden and distract the learner's attention away from the video instruction (Bellini & Akullian, 2007; Rayner, Denholm, & Sigafoos, 2009). However, research has also shown that participants with ASD and intellectual disabilities require additional error-correction procedures to improve their overall learning through video modelling (Banda et al., 2010; Copple et al., 2015). The use of additional instruction (e.g., a systematic prompting system) was deemed essential in maintaining engagement to the task and learning the navigational pathways to the required vocabulary, due to the participant learning characteristics associated with intellectual disabilities.

The degree of prompting required for each participant differed and could have been influenced by the severity of ASD and intellectual disabilities, as well as their level of preoccupation, prompt-dependence, and communicative competence. No data were collected regarding these specific participant characteristics. Therefore, the information that

follows is based on observations within sessions. All participants exhibited varying levels of preoccupation, which appeared to have substantially affected their independent conversation behavior. Furthermore, Dan also demonstrated higher levels of prompt dependence than others and often waited for a cue before responding (e.g., he generally held his finger over the communication symbol and looked at the researcher for confirmation before activating the symbols on his communication device). Given that Dan was much older (18 years) than the others (11 years), it is possible that he had experienced longer exposure to stimulus control instructions before the start of the study and had become more conditioned to prompting compared to the others. Although all participants were able to request independently, they lacked operational competence to navigate through the AAC system to demonstrate other communicative behaviors. This resulted in the use of navigational and gestural prompts to teach the pathways for the required vocabulary.

Clinical implications

The findings of this study underline the following clinical implications: first, the detailed protocols for developing VBI that were developed for the current investigation can be used and modified by relevant stakeholders to facilitate skill development in individuals with ASD who use AAC. Second, educators might need to consider various strategies in maintaining participant engagement with the video clips once the novelty wears off. For example, it might be necessary to reinforce attention to the video clips or use a range of models to maintain engagement. Third, the use of peers or adults as models during the video-instruction might be more time and resource efficient. This investigation showed that creating video models using peers as models was more time efficient than self-modelling clips (i.e., 30 min vs. 90 min), and that peer models were also effective in promoting skill development. Fourth, practitioners must understand the implementation of different types of SI (e.g., progressive time-delay, most-to-least prompts, graduated guidance), and tailor the type of SI for use in combination with VBI based on the learning and communication needs of each individual (McMillan & Renzaglia, 2014). The findings from the current study showed that while systematic instruction was an important component of the intervention, the system of least prompts might not have been the most suitable for all participants. For example, Dan and Lily needed more intrusive prompts and a longer intervention period and so might have benefitted from methods such as progressive time-delay or most-to-least prompting.

Finally, this study highlighted the importance of a backup AAC system for students, in the event of a device malfunction or absence. One of the main challenges encountered in this study was the absence or dysfunction of the SGD (i.e., flat battery). Although the school had similar apps in their school iPad,³ the system was set-up using a generic

³The iPad is a series of tablet computers that runs on the iOS mobile operating system and is designed by Apple Inc., Cupertino, California, <https://www.apple.com/au/ipad/>

core vocabulary template. Hence, the participants who used the core vocabulary template in their personal SGD were able to use the school AAC system, but the participants who used customized templates were unable to do so.

Limitations and future directions

Although the investigation met many of the quality markers for single-case experimental research design (Kratochwill et al., 2010; Reichow & Volkmar, 2010), the findings of this investigation should be considered within the context of its limitations. First, it was difficult to extrapolate which component of the intervention – the video-based modelling intervention or the system of least prompts – yielded more impact on the treatment outcome. Second, challenges encountered during participant recruitment resulted in fewer participants than envisaged (i.e., each strand of the investigation included two participants). Although two participants are technically sufficient to demonstrate treatment effect, replication across more than two units (across participants, settings, or behaviors) is preferred when demonstrating an experimental effect (Kazdin, 2011; Kratochwill et al., 2010). Third, there was a mismatch between the expected and demonstrated communicative competence by the participants. All the participants were deemed to be context-dependent communicators by their teachers. However, given the broad definition and continuum of context-dependent communicators, there was high variability in the observed communication abilities of the participants. Finally, this investigation was conducted in a separate controlled setting, as opposed to a naturalistic setting, which is common in interventions for adolescents with ASD (Holyfield et al., 2017). While implementing interventions in naturalistic settings for children with ASD is consistent with recommended practice, little is known about the evidence for conducting intervention for adolescents and adults with ASD in naturalistic settings (Reichow & Volkmar, 2010).

This was an initial, exploratory study that investigated the use of VM and VSM in teaching conversation behaviors to individuals with ASD, who use AAC. Additional intervention studies replicating similar protocols are required to obtain sufficient evidence for the use of VBI to teach conversational behaviors to individuals with ASD, who use AAC. Research is also needed to study the use of PoVM for this group. Unlike natural speakers, individuals who use AAC require additional unique skills, such as understanding the navigational pathways to access appropriate vocabulary, as well as understanding the symbols used in AAC systems (Light & McNaughton, 2014). All participants in this investigation required additional instruction to learn the navigation pathways to access vocabulary and because PoVM is presented from the vantage point of the learner, it might facilitate the acquisition of navigational competence. Given the potential impact of systematic instruction combined with VBI on the acquisition of conversation behavior for individuals with ASD who use AAC, future research should evaluate communication partner training. Specifically, future research should evaluate the implementation of SI by communication

partners based on the learning and communication needs of individuals.

Conclusion

The current investigation offers an initial framework for implementing video-based interventions to develop conversational behavior in individuals with ASD. It seems evident that individuals with ASD face numerous challenges engaging socially, and that far greater challenges exist for individuals with ASD who have complex communication needs. Findings from the current investigation and previous research (Banda et al., 2010; Copple et al., 2015) show that for individuals with ASD who also have intellectual disabilities, video-based intervention is most effective when implemented with systematic instruction to facilitate the acquisition of conversational behaviors.

Author note

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Disclosure statement

No potential conflict of interest was reported by the authors.

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